Software Requirements Specification (SRS) for CogniWallet

1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for CogniWallet, a comprehensive mobile wallet system. This document provides a detailed description of the functionalities, constraints, and system behavior to serve as a foundation for the development, testing, and deployment of CogniWallet.

1.2 Scope

CogniWallet is a secure, scalable, and feature-rich mobile wallet system designed to facilitate digital transactions, bill payments, fund transfers, and financial management. It aims to serve individuals and businesses by offering a user-friendly platform integrated with advanced security measures, multi-currency support, and regulatory compliance. The system will support seamless integration with merchants, banks, and third-party APIs while delivering an exceptional user experience.

1.3 Definitions, Acronyms, and Abbreviations

- **KYC**: Know Your Customer – a process to verify the identity of users.
- **AML**: Anti-Money Laundering – regulatory frameworks to prevent money laundering.
- **GDPR**: General Data Protection Regulation – a European Union regulation for data protection and privacy.
- **PIN**: Personal Identification Number – a security code for authentication.
- **UI/UX**: User Interface/User Experience – the design and usability of the system interface.

1.4 References

1. ISO/IEC 27001: Information Security Management Standards
2. OWASP Mobile Security Testing Guide
3. Relevant financial regulations such as PSD2, GDPR, and AML guidelines
4. Industry best practices for mobile wallet systems

1.5 Overview

The document is structured as follows:

- Section 2: Overall Description
- Section 3: Specific Requirements
- Section 4: System Features

- Section 5: Non-Functional Requirements
- Section 6: Assumptions and Dependencies
- Section 7: Appendices

2. Overall Description

2.1 Product Perspective

CogniWallet will function as an independent, standalone application with optional integrations into banking systems, merchant platforms, and third-party APIs. The system will employ a microservices-based architecture to ensure scalability, modularity, and ease of maintenance. The following components define its ecosystem:

- **Mobile Application**: The primary user interface for customers to perform transactions, manage finances, and access features.
- **Backend Services**: A secure API-driven service layer that handles transaction processing, user authentication, and data storage.
- **Admin Portal**: A web-based platform for system administrators to manage users, monitor transactions, and configure settings.
- **Integration Layer**: APIs and webhooks for connecting with external services, such as banks, merchants, and regulatory bodies.
- **Notification Service**: Real-time notifications delivered via push, SMS, and email.

System Interfaces:

- **External APIs**: Banking APIs for fund transfers, exchange rate providers for multi-currency support, and utility provider APIs for bill payments.
- **User Devices**: iOS and Android platforms for mobile applications.
- **Admin Tools**: Web-based administrative dashboards for backend management.

2.2 Product Functions

CogniWallet will support the following core functions:

1. **User Registration and Authentication**:
    - Multi-factor authentication, including biometric and PIN-based login.
    - User onboarding through KYC compliance and document upload.
2. **Digital Payments**:
    - QR code payments, NFC payments, and peer-to-peer transfers.
3. **Fund Management**:
    - Linking and managing bank accounts, credit/debit cards, and wallet balance.
    - Real-time balance updates and multi-currency management.

4. **Financial Tools**:
    - Expense tracking and categorization.
    - Savings goals and financial insights.
5. **Merchant Transactions**:
    - Payment gateway for businesses to accept digital payments.
6. **Notifications and Alerts**:
    - Transaction alerts, reminders, and promotional notifications.

2.3 User Classes and Characteristics

- **End Users**:
    - Individuals using the wallet for personal financial needs.
    - Small business owners using the wallet for transactions and payments.
    - Users with varying levels of tech literacy.
- **Administrators**:
    - System administrators managing backend operations, user accounts, and transactions.
- **Merchants**:
    - Retailers and service providers using CogniWallet for accepting payments.

2.4 Operating Environment

- **Mobile Application**:
    - Platforms: Android (minimum version 10), iOS (minimum version 13).
    - Devices: Smartphones and tablets.
- **Backend Systems**:
    - Cloud-hosted with 99.99% availability, leveraging Kubernetes for container orchestration.
    - Database: PostgreSQL for relational data and MongoDB for NoSQL storage.
- **Network**:
    - Minimum 4G connectivity for seamless user experience.
    - Offline capabilities for some features (e.g., QR code generation).

2.5 Constraints

- Compliance with financial regulations (e.g., PSD2, GDPR, AML).
- Latency of under 500 milliseconds for transaction-related APIs.
- Data localization requirements for specific regions.

2.6 Assumptions and Dependencies

- Users will have smartphones with biometric authentication capabilities.
- Regulatory approvals will be obtained for each region before deployment.
- Partnerships with banks and utility providers will be established for integrations.

3. Specific Requirements

3.1 Functional Requirements

3.1.1 User Registration and Authentication

1. Users must be able to register using their mobile number or email address.
2. The system must support KYC verification, requiring users to upload identity documents and complete a verification process.
3. Multi-factor authentication (MFA) must be implemented, with options for:
    - Biometric authentication (fingerprint or facial recognition).
    - PIN-based authentication.
4. Password recovery should be enabled via email or mobile number OTP.

3.1.2 Wallet Management

1. Users must be able to add and manage multiple funding sources, including:
    - Bank accounts.
    - Credit or debit cards.
    - Other mobile wallets.
2. The system should display the wallet balance in real time.
3. Multi-currency wallets must:
    - Display balances in different currencies.
    - Support manual and automatic currency conversion at live exchange rates.

3.1.3 Digital Payments

1. Users should be able to make payments via:
    - QR code scanning.
    - NFC (Near Field Communication).
    - Phone number or wallet ID.
2. Peer-to-peer transfers must:
    - Allow users to send money to contacts in their phonebook.
    - Support scheduled or recurring transfers.
3. The system must generate unique payment request IDs for tracking.

3.1.4 Bill Payments and Utilities

1. The system must integrate with service providers to allow:
    - Utility bill payments (e.g., electricity, water, internet).
    - Mobile recharge and top-ups.
2. Users should be able to view payment history for all bills.

3.1.5 Expense Tracking and Financial Insights

1. The system must categorize transactions automatically based on predefined rules (e.g., groceries, dining, transportation).
2. Users should be able to:
    - View expense summaries in graphical formats.
    - Set monthly spending limits with alerts for threshold breaches.
3. Savings tools must allow users to:
    - Create and track savings goals.
    - Automate savings by setting rules (e.g., save 10% of income).

3.1.6 Merchant Transactions

1. Merchants must be able to generate payment links or QR codes for receiving payments.
2. The system should provide merchants with a transaction history dashboard, including:
    - Daily, weekly, and monthly sales summaries.
    - Refund processing options.

3.1.7 Notifications

1. Users must receive real-time notifications for:
    - Successful and failed transactions.
    - Incoming payments and refunds.
    - System updates and promotional offers.
2. Notifications should be customizable, allowing users to choose between:
    - Push notifications.
    - SMS.
    - Email.

3.1.8 Security and Fraud Prevention

1. End-to-end encryption must be implemented for all communications.
2. Fraud detection algorithms must:
   - Flag suspicious activity (e.g., repeated failed login attempts, large transactions).
   - Notify users and administrators of flagged events.
3. Users should have the ability to lock or disable their wallet in case of suspicious activity.

3.2 Data Management Requirements

1. User data must be encrypted at rest and in transit.
2. All transaction data must be stored for a minimum of five years to comply with regulatory requirements.
3. The system must implement role-based access control (RBAC) for administrators.
4. Data localization rules should be adhered to for regions where applicable.

3.3 Integration Requirements

1. The system must integrate with:
   - Payment gateways for processing transactions.
   - Banks and financial institutions for fund transfers.
   - Exchange rate APIs for live currency conversion.
2. Open APIs must be provided for third-party integrations, including:
   - Merchant platforms.
   - Loyalty programs.

3.4 Performance Requirements

1. Transaction processing time must not exceed 500 milliseconds under normal load.
2. The system must handle:
   - Up to 100,000 concurrent users.
   - Peak transaction volumes of 1,000 TPS (transactions per second).
3. The mobile application must load within 3 seconds on supported devices.

3.5 Usability Requirements

1. The mobile application must support:
   - Gesture-based navigation.
   - Voice commands for specific actions (e.g., "Send $50 to John").
2. Accessibility features should include:
   - Support for screen readers.

- High-contrast modes for visually impaired users.

3.6 Compliance Requirements

1. The system must comply with:
    - GDPR for data protection and privacy.
    - PSD2 for secure digital payment processing.
    - AML regulations to prevent illicit transactions.
2. All user data must be anonymized where possible to enhance privacy.

4. System Features

This section provides an in-depth description of the features identified in Section 3. Each feature includes a detailed explanation of its purpose, functionality, and interactions with other features.

4.1 User Registration and Authentication

- **Purpose**: To securely onboard users and ensure only authorized access to the system.
- **Features**:
    - **Account Creation**: Users can register using their mobile number or email address and verify their identity through KYC processes.
    - **Biometric Authentication**: Login options include fingerprint and facial recognition.
    - **Multi-Factor Authentication (MFA)**:
        - OTP (One-Time Password) via SMS or email.
        - PIN setup for additional security.
    - **Forgot Password**: Users can recover accounts securely through OTP verification and reset options.

4.2 Wallet Management

- **Purpose**: To enable users to manage their funds efficiently within a secure digital wallet.
- **Features**:
    - **Wallet Overview**:
        - Real-time display of wallet balance.
        - Breakdowns by linked accounts (e.g., bank, credit cards).
    - **Multi-Currency Support**:
        - Wallets can hold balances in multiple currencies.
        - Real-time exchange rates are displayed for conversions.

    - **Funding Sources**:
        - Add, remove, or manage bank accounts and cards.
        - Direct transfers between wallet and external accounts.

4.3 Digital Payments

- **Purpose**: To facilitate seamless transactions between users, merchants, and service providers.
- **Features**:
    - **Peer-to-Peer Transfers**:
        - Transfer funds via phone number, wallet ID, or QR code.
        - Schedule recurring payments with notifications.
    - **Merchant Payments**:
        - Scan merchant QR codes or use NFC for instant payments.
    - **Transaction Tracking**:
        - Unique transaction IDs for easy tracking.
        - Transaction history with filters for date, amount, and recipient.

4.4 Bill Payments and Utilities

- **Purpose**: To allow users to pay bills and manage utilities directly within the app.
- **Features**:
    - **Bill Categories**:
        - Electricity, water, internet, subscriptions, and mobile recharges.
    - **Payment History**:
        - Detailed records of paid bills with downloadable receipts.
    - **Reminders**:
        - Notifications for upcoming due dates and missed payments.

4.5 Expense Tracking and Financial Insights

- **Purpose**: To help users manage their finances and achieve their savings goals.
- **Features**:
    - **Transaction Categorization**:
        - Automatic grouping of expenses (e.g., groceries, dining, travel).
    - **Budget Management**:
        - Users can set monthly budgets for different categories.
        - Alerts are sent if spending exceeds predefined limits.
    - **Savings Tools**:
        - Create savings goals with progress tracking.

            - Automate savings through rules (e.g., round-ups, percentage of income).

4.6 Merchant Transactions

- **Purpose**: To support businesses in receiving payments and managing finances.
- **Features**:
    - **Payment Requests**:
        - Merchants can generate and share payment links or QR codes.
    - **Transaction Dashboard**:
        - Visual summaries of sales, refunds, and disputes.
    - **Refunds and Adjustments**:
        - Merchants can initiate refunds with user notifications.

4.7 Notifications

- **Purpose**: To keep users informed about important events and updates in real-time.
- **Features**:
    - **Customizable Alerts**:
        - Users can select preferred notification channels (push, SMS, or email).
    - **Event Types**:
        - Transaction confirmations, promotions, security alerts, and reminders.
    - **In-App Notifications**:
        - Centralized notification hub for retrievable messages.

4.8 Security and Fraud Prevention

- **Purpose**: To ensure system integrity and protect user data and funds from unauthorized access.
- **Features**:
    - **Real-Time Fraud Detection**:
        - AI-driven algorithms to identify and flag suspicious activities.
    - **User-Controlled Wallet Lock**:
        - Users can lock their accounts temporarily in case of suspicious activity.
    - **Data Encryption**:
        - All sensitive data is encrypted at rest and during transmission.

4.9 Integration with External Systems

- **Purpose**: To extend the system's functionality through seamless integration with third-party services.
- **Features**:
  - **Bank Integrations**:
    - Direct bank transfers and real-time balance synchronization.
  - **Utility Providers**:
    - Support for bill payments and subscription renewals.
  - **Open APIs**:
    - Allow third-party developers to build tools that integrate with CogniWallet.

4.10 Accessibility and Localization

- **Purpose**: To ensure usability across diverse user demographics and regions.
- **Features**:
  - **Multi-Language Support**:
    - Interface available in multiple languages, with region-specific settings.
  - **Accessibility Tools**:
    - Support for screen readers, voice navigation, and high-contrast modes.
  - **Localized Services**:
    - Region-specific utilities and currency handling.

4.11 Reporting and Analytics

- **Purpose**: To provide users and administrators with insights into financial activity.
- **Features**:
  - **User Reports**:
    - Monthly and yearly summaries of expenses, savings, and income.
  - **Admin Dashboards**:
    - Real-time monitoring of system usage, transactions, and security events.

5. Non-Functional Requirements

This section outlines the system's quality attributes, constraints, and operational parameters that support the functional requirements described earlier.

5.1 Performance Requirements

1. **Transaction Latency**: The system must process transactions within 500 milliseconds under normal load conditions.
2. **Concurrent Users**: The system must support up to 100,000 concurrent users without performance degradation.

3. **Peak Transaction Volume**: The system should handle up to 1,000 transactions per second (TPS) during peak loads.

5.2 Scalability Requirements

1. **Horizontal Scaling**: The system architecture must support the addition of more servers or instances to handle growth in user base and transaction volume.

5.3 Security Requirements

1. **Data Encryption**:
   - All data in transit must be encrypted using HTTPS and TLS protocols.
   - All data at rest must be encrypted using AES-256 encryption.
2. **Authentication**:
   - Implement multi-factor authentication (MFA) for both end users and administrators.
3. **Fraud Prevention**:
   - Real-time fraud detection and prevention mechanisms must be in place.
4. **Role-Based Access Control (RBAC)**:
   - System components and user data must be accessible only to authorized personnel based on roles and permissions.
5. **Session Management**:
   - Implement secure session handling with timeouts and token-based authentication.

5.4 Availability Requirements

1. **Uptime**:
   - The system must ensure 99.99% availability, including planned maintenance.
2. **Disaster Recovery**:
   - A disaster recovery plan must be implemented with data backup intervals not exceeding 5 minutes.
   - The system must have an RTO (Recovery Time Objective) of 30 minutes and an RPO (Recovery Point Objective) of 5 minutes.

5.5 Usability Requirements

1. **Accessibility**:

- The system must comply with accessibility standards, such as WCAG 2.1, to support users with disabilities.

2. **Mobile Optimization**:
   - The application must provide a consistent and intuitive user experience across all supported devices.
3. **Loading Time**:
   - The mobile application must load within 3 seconds on average internet connections (4G or higher).

5.6 Compliance Requirements

1. **Regulatory Compliance**:
   - Adhere to GDPR, PSD2, AML, and other relevant regional regulations.
2. **Audit Trail**:
   - All user and system activities must be logged with a timestamp and retained for at least 5 years for regulatory compliance.
3. **Data Localization**:
   - User data must be stored within the geographical boundaries of regions requiring data localization.

5.7 Maintainability Requirements

1. **Code Modularity**:
   - The system must use a modular code structure to facilitate easy maintenance and updates.
2. **Documentation**:
   - Comprehensive technical documentation must be provided for developers, administrators, and end users.
3. **Testing**:
   - Automated unit, integration, and regression tests must cover at least 90% of the codebase.

5.8 Interoperability Requirements

1. **API Standards**:
   - The system must support OpenAPI standards for RESTful APIs to enable third-party integrations.
2. **External System Integration**:
   - Provide adapters for seamless interaction with banking systems, payment gateways, and utility services.

5.9 Localization Requirements

1. **Language Support**:
   - The application must support at least 10 regional languages at launch.
2. **Time Zone Support**:
   - All timestamps must be adjusted according to the user's local time zone.

5.10 Environmental Requirements

1. **Operating Systems**:
   - Mobile application must support Android (minimum version 10) and iOS (minimum version 13).
2. **Network Requirements**:
   - The application should function optimally on 4G or higher connections, with basic functionality on 3G.

6. Assumptions and Dependencies

This section outlines the assumptions made during the development of CogniWallet and the dependencies required for its successful operation.

6.1 Assumptions

1. **User Device Capabilities**:
   - Users will have smartphones with biometric authentication (fingerprint or facial recognition).
   - Users will have devices running at least Android 10 or iOS 13.
2. **Network Availability**:
   - Users will have access to reliable internet connections (4G or higher) for optimal performance.
3. **Regulatory Compliance**:
   - Necessary regulatory approvals (e.g., PSD2, GDPR, AML) will be granted in regions of operation.
4. **Partnerships**:
   - Partnerships with banks, utility providers, and payment gateways will be established prior to the system's launch.
5. **User Literacy**:
   - Users will have basic digital literacy to navigate the app and perform transactions.

6.2 Dependencies

1. **Third-Party Services**:
   - Payment gateways for processing transactions.
   - Bank APIs for fund transfers and account synchronization.
   - Exchange rate providers for multi-currency functionality.
2. **Cloud Infrastructure**:
   - The system will depend on a cloud hosting provider, such as AWS, Azure, or Google Cloud, for backend services and databases.
3. **Regulatory Authorities**:
   - Compliance requirements, including KYC, AML, and data localization, depend on the cooperation of regulatory authorities.
4. **Technology Stack**:
   - The system relies on specific frameworks and technologies, including:
     - Kubernetes for container orchestration.
     - PostgreSQL for relational database management.
     - MongoDB for NoSQL storage.
5. **Development and Testing Tools**:
   - Dependency on CI/CD pipelines for automated testing and deployment.
   - Tools for performance monitoring and logging (e.g., Grafana, Prometheus).

7. Appendices

This section provides additional information, definitions, and references that support the Software Requirements Specification for CogniWallet.

7.1 Glossary

- **KYC (Know Your Customer)**: A process by which user identities are verified for compliance with financial regulations.
- **AML (Anti-Money Laundering)**: A regulatory framework to prevent illegal money laundering activities.
- **GDPR (General Data Protection Regulation)**: A European Union regulation governing data protection and privacy.
- **TPS (Transactions Per Second)**: A measure of how many transactions a system can process in one second.
- **RBAC (Role-Based Access Control)**: A security mechanism for controlling access to resources based on user roles.

- **MFA (Multi-Factor Authentication)**: A security process that requires multiple forms of verification to access the system.

7.2 Acronyms

- **API**: Application Programming Interface.
- **UI/UX**: User Interface/User Experience.
- **OTP**: One-Time Password.
- **RTO**: Recovery Time Objective.
- **RPO**: Recovery Point Objective.

7.3 References

1. ISO/IEC 27001: Standards for Information Security Management Systems.
2. OWASP Mobile Security Testing Guide: Best practices for securing mobile applications.
3. Relevant financial regulations, including:
    - PSD2 (Payment Services Directive 2).
    - GDPR (General Data Protection Regulation).
    - AML (Anti-Money Laundering) guidelines.
4. Industry reports on mobile wallet trends and usage.